

Elastic Net regression

ml_dataset/data/model_ready/movie_success_v6/colab_runs_v2/results_elastic_net.json

```

1 {
2   "model": "elastic_net",
3   "feature_count": 360,
4   "best_params_imdb": {
5     "model_alpha": 1.0,
6     "model_l1_ratio": 0.05
7   },
8   "best_params_wom": {
9     "model_alpha": 0.1778279410038923,
10    "model_l1_ratio": 0.85
11  },
12  "metrics_real_imdb_baseline": {
13    "n": 10,
14    "r2": -0.2708253025366527,
15    "mae": 0.7007201732778838,
16    "rmse": 0.7849988466233709,
17    "spearman": -0.7500139429824981,
18    "pearson": -0.7791196785574992
19  },
20  "metrics_real_wom_baseline": {
21    "n": 10,
22    "r2": -0.2478920091782968,
23    "mae": 0.5776290508955284,
24    "rmse": 0.6750334811442039,
25    "spearman": -0.9636363636363635,
26    "pearson": -0.981368399516436
27  },
28  "metrics_real_imdb_tuned": {
29    "n": 10,
30    "r2": 0.11503526291994615,
31    "mae": 0.5331430752502533,
32    "rmse": 0.6550720578761683,
33    "spearman": 0.46342324932251916,
34    "pearson": 0.4711022117393806
35  },
36  "metrics_real_wom_tuned": {
37    "n": 10,
38    "r2": 0.4314474940583273,
39    "mae": 0.3551657594246886,
40    "rmse": 0.45564027069481433,
41    "spearman": 0.6848484848484848,
42    "pearson": 0.7426719953302064
43  },
44  "metrics_aug_imdb_baseline": {
45    "n": 50,
46    "r2": -0.013460327455919296,
47    "mae": 0.4998999999999998,
48    "rmse": 0.5785758377257038,
49    "spearman": -0.1708965015387121,
50    "pearson": -0.1546915351037204
51  },
52  "metrics_aug_wom_baseline": {
53    "n": 50,
54    "r2": -0.02814640834228621,
55    "mae": 0.46040510000000007,
56    "rmse": 0.5559736496791012,
57    "spearman": -0.21853819706731129,
58    "pearson": -0.22369188468778975
59  },
60  "metrics_aug_imdb_tuned": {
61    "n": 50,
62    "r2": 0.3648764581904227,
63    "mae": 0.3474594729158694,
64    "rmse": 0.45802166581273274,
65    "spearman": 0.5886245714861638,
66    "pearson": 0.6060286833274762
67  },
68  "metrics_aug_wom_tuned": {
69    "n": 50,
70    "r2": 0.13077583394046244,
71    "mae": 0.40598145605296454,

```

```
72  "rmse": 0.5112020931132174,  
73  "spearman": 0.3786314525810324,  
74  "pearson": 0.36338785641158156  
75  }  
76 }
```

Ridge regression

ml_dataset/data/model_ready/movie_success_v6/colab_runs_v2/results_ridge.json

```

1 {
2   "model": "ridge",
3   "feature_count": 360,
4   "best_params_imdb": {
5     "model_alpha": 117.21022975334793
6   },
7   "best_params_wom": {
8     "model_alpha": 417.53189365604004
9   },
10  "metrics_real_imdb_baseline": {
11    "n": 10,
12    "r2": -0.9728363192347182,
13    "mae": 0.8978673293065762,
14    "rmse": 0.9780737861720428,
15    "spearman": -0.7134278969833519,
16    "pearson": -0.677824374372566
17  },
18  "metrics_real_wom_baseline": {
19    "n": 10,
20    "r2": -0.7593256474224326,
21    "mae": 0.7180099488395053,
22    "rmse": 0.8015118158937188,
23    "spearman": -0.7696969696969697,
24    "pearson": -0.7600424755485007
25  },
26  "metrics_real_imdb_tuned": {
27    "n": 10,
28    "r2": -0.2601869951946112,
29    "mae": 0.6853429056306855,
30    "rmse": 0.7817062581109779,
31    "spearman": -0.9695302189773756,
32    "pearson": -0.9911926454654585
33  },
34  "metrics_real_wom_tuned": {
35    "n": 10,
36    "r2": -0.25448353131909385,
37    "mae": 0.577648746314265,
38    "rmse": 0.6768139388557993,
39    "spearman": -0.9999999999999999,
40    "pearson": -0.9959681259428614
41  },
42  "metrics_aug_imdb_baseline": {
43    "n": 50,
44    "r2": 0.37957119455356414,
45    "mae": 0.3541086124863018,
46    "rmse": 0.4526920765312549,
47    "spearman": 0.6322745172255105,
48    "pearson": 0.6462947869702826
49  },
50  "metrics_aug_wom_baseline": {
51    "n": 50,
52    "r2": 0.05084032554564666,
53    "mae": 0.40638863633191413,
54    "rmse": 0.5341907538996831,
55    "spearman": 0.3467466986794718,
56    "pearson": 0.36894160059257514
57  },
58  "metrics_aug_imdb_tuned": {
59    "n": 50,
60    "r2": 0.471567091285498,
61    "mae": 0.3172082436744603,
62    "rmse": 0.41778403928349733,
63    "spearman": 0.667598729102777,
64    "pearson": 0.6871415621114381
65  },
66  "metrics_aug_wom_tuned": {
67    "n": 50,
68    "r2": 0.14659387138511237,
69    "mae": 0.40598175331583475,
70    "rmse": 0.5065293396716725,
71    "spearman": 0.3581752701080432,
72    "pearson": 0.385210790250144

```

73 }
74 }

Random Forest regression

ml_dataset/data/model_ready/movie_success_v6/colab_runs_v2/results_random_forest.json

```

1 {
2   "model": "random_forest",
3   "feature_count": 360,
4   "best_params_imdb": {
5     "model_max_depth": 7,
6     "model_max_features": 0.2,
7     "model_min_samples_leaf": 1,
8     "model_min_samples_split": 2,
9     "model_n_estimators": 200
10  },
11  "best_params_wom": {
12    "model_max_depth": 10,
13    "model_max_features": 0.2,
14    "model_min_samples_leaf": 1,
15    "model_min_samples_split": 2,
16    "model_n_estimators": 200
17  },
18  "metrics_real_imdb_baseline": {
19    "n": 10,
20    "r2": -0.4582073737998693,
21    "mae": 0.7341333333333342,
22    "rmse": 0.8408833186331839,
23    "spearman": -0.8414790579803637,
24    "pearson": -0.8297102931322186
25  },
26  "metrics_real_wom_baseline": {
27    "n": 10,
28    "r2": -0.3780077542913005,
29    "mae": 0.6143983000000004,
30    "rmse": 0.7093533807154161,
31    "spearman": -0.6363636363636362,
32    "pearson": -0.7465520282498894
33  },
34  "metrics_real_imdb_tuned": {
35    "n": 10,
36    "r2": -0.2519898780249983,
37    "mae": 0.6850357777777771,
38    "rmse": 0.7791597344924349,
39    "spearman": -0.9939209163101398,
40    "pearson": -0.9987764152990641
41  },
42  "metrics_real_wom_tuned": {
43    "n": 10,
44    "r2": -0.2441355193020549,
45    "mae": 0.5697737963888893,
46    "rmse": 0.6740166993938812,
47    "spearman": -0.9878787878787878,
48    "pearson": -0.9772030696900543
49  },
50  "metrics_aug_imdb_baseline": {
51    "n": 50,
52    "r2": 0.4984078774569972,
53    "mae": 0.3193733333333334,
54    "rmse": 0.4070354830287453,
55    "spearman": 0.6781863675401048,
56    "pearson": 0.7483924710240085
57  },
58  "metrics_aug_wom_baseline": {
59    "n": 50,
60    "r2": 0.19770551997375818,
61    "mae": 0.39171838666666686,
62    "rmse": 0.49112679537932874,
63    "spearman": 0.4020648259303721,
64    "pearson": 0.4593517585863801
65  },
66  "metrics_aug_imdb_tuned": {
67    "n": 50,
68    "r2": 0.5440413305775964,
69    "mae": 0.29916571428571437,
70    "rmse": 0.3880785646552739,
71    "spearman": 0.6831914329832051,
72    "pearson": 0.752835300233795

```

```
73 },
74 "metrics_aug_wom_tuned": {
75   "n": 50,
76   "r2": 0.20747369193279486,
77   "mae": 0.38121092833333337,
78   "rmse": 0.48812783230452844,
79   "spearman": 0.41512605042016804,
80   "pearson": 0.4577009889721229
81 }
82 }
```

Gradient Boosting regression

ml_dataset/data/model_ready/movie_success_v6/colab_runs_v2/results_gradient_boosting.json

```

1 {
2   "model": "gradient_boosting",
3   "feature_count": 360,
4   "best_params_imdb": {
5     "model_l2_regularization": 10.0,
6     "model_learning_rate": 0.05,
7     "model_max_depth": 3,
8     "model_max_iter": 100,
9     "model_max_leaf_nodes": 15,
10    "model_min_samples_leaf": 4
11  },
12  "best_params_wom": {
13    "model_l2_regularization": 1.0,
14    "model_learning_rate": 0.1,
15    "model_max_depth": 3,
16    "model_max_iter": 800,
17    "model_max_leaf_nodes": 15,
18    "model_min_samples_leaf": 10
19  },
20  "metrics_real_imdb_baseline": {
21    "n": 10,
22    "r2": -0.23456790123456805,
23    "mae": 0.6777777777777778,
24    "rmse": 0.7737195714912748,
25    "spearman": -1.0,
26    "pearson": -0.9999999999999994
27  },
28  "metrics_real_wom_baseline": {
29    "n": 10,
30    "r2": -0.23456790123456783,
31    "mae": 0.5727511111111111,
32    "rmse": 0.6714200448800041,
33    "spearman": -0.9999999999999999,
34    "pearson": -1.0
35  },
36  "metrics_real_imdb_tuned": {
37    "n": 10,
38    "r2": -0.2251716862483224,
39    "mae": 0.678760719910359,
40    "rmse": 0.7707695833787238,
41    "spearman": -0.8963581269790831,
42    "pearson": -0.9475436646211018
43  },
44  "metrics_real_wom_tuned": {
45    "n": 10,
46    "r2": 0.2454007071815033,
47    "mae": 0.4464603611186936,
48    "rmse": 0.5249222831208017,
49    "spearman": 0.5515151515151515,
50    "pearson": 0.5098777107770938
51  },
52  "metrics_aug_imdb_baseline": {
53    "n": 50,
54    "r2": 0.27796173456051754,
55    "mae": 0.3620064228004976,
56    "rmse": 0.4883565574738634,
57    "spearman": 0.5423277161374851,
58    "pearson": 0.5403484325646949
59  },
60  "metrics_aug_wom_baseline": {
61    "n": 50,
62    "r2": -0.03237681070674259,
63    "mae": 0.4453785735193426,
64    "rmse": 0.5571162777211551,
65    "spearman": 0.18559423769507805,
66    "pearson": 0.21589046813079724
67  },
68  "metrics_aug_imdb_tuned": {
69    "n": 50,
70    "r2": 0.476267632315603,
71    "mae": 0.3065715287935003,
72    "rmse": 0.4159217426098653,

```

```
73     "spearman": 0.7115374286176872,  
74     "pearson": 0.6902292245579901  
75 },  
76 "metrics_aug_wom_tuned": {  
77     "n": 50,  
78     "r2": 0.13603401663449433,  
79     "mae": 0.39415854743339457,  
80     "rmse": 0.5096535446746526,  
81     "spearman": 0.36393757503001195,  
82     "pearson": 0.3961836732314205  
83 }  
84 }
```


Partial Least Squares regression

ml_dataset/data/model_ready/movie_success_v6/colab_runs_v2/results_pls.json

```

1 {
2   "model": "pls",
3   "feature_count": 360,
4   "best_params_imdb": {
5     "model_n_components": 2
6   },
7   "best_params_wom": {
8     "model_n_components": 1
9   },
10  "metrics_real_imdb_baseline": {
11    "n": 10,
12    "r2": -1.0449645499206999,
13    "mae": 0.9144580683603671,
14    "rmse": 0.9957928048828969,
15    "spearman": -0.7012325483169698,
16    "pearson": -0.6826439226268518
17  },
18  "metrics_real_wom_baseline": {
19    "n": 10,
20    "r2": -0.8464575428277019,
21    "mae": 0.7398677378241273,
22    "rmse": 0.8211197065838755,
23    "spearman": -0.7696969696969697,
24    "pearson": -0.7098074857507141
25  },
26  "metrics_real_imdb_tuned": {
27    "n": 10,
28    "r2": -0.9746837582436303,
29    "mae": 0.8984793665010258,
30    "rmse": 0.9785316317689153,
31    "spearman": -0.7134278969833519,
32    "pearson": -0.6772474717248187
33  },
34  "metrics_real_wom_tuned": {
35    "n": 10,
36    "r2": -0.8658732412937531,
37    "mae": 0.7112729107685098,
38    "rmse": 0.8254254979692728,
39    "spearman": -0.7818181818181817,
40    "pearson": -0.7596015607465906
41  },
42  "metrics_aug_imdb_baseline": {
43    "n": 50,
44    "r2": 0.47192330845360564,
45    "mae": 0.3087251093119983,
46    "rmse": 0.4176432012191031,
47    "spearman": 0.6802557695983097,
48    "pearson": 0.6891619239696984
49  },
50  "metrics_aug_wom_baseline": {
51    "n": 50,
52    "r2": -0.029577368263506765,
53    "mae": 0.4354062397875334,
54    "rmse": 0.5563604133627774,
55    "spearman": 0.2590636254501801,
56    "pearson": 0.27789534325288023
57  },
58  "metrics_aug_imdb_tuned": {
59    "n": 50,
60    "r2": 0.47192330845360564,
61    "mae": 0.3087251093119983,
62    "rmse": 0.4176432012191031,
63    "spearman": 0.6802557695983097,
64    "pearson": 0.6891619239696984
65  },
66  "metrics_aug_wom_tuned": {
67    "n": 50,
68    "r2": 0.1897278534737039,
69    "mae": 0.4020334360923966,
70    "rmse": 0.4935625307425964,
71    "spearman": 0.38525810324129645,
72    "pearson": 0.46090226088852276

```

73 }
74 }

Gaussian Process regression

ml_dataset/data/model_ready/movie_success_v6/colab_runs_v2/results_gaussian_process.json

```

1 {
2   "model": "gaussian_process",
3   "feature_count": 360,
4   "best_params_imdb": {
5     "model_kernel": "iso_rbf"
6   },
7   "best_params_wom": {
8     "model_kernel": "iso_rbf"
9   },
10  "metrics_real_imdb_baseline": {
11    "n": 10,
12    "r2": -0.23456790123456805,
13    "mae": 0.6777777777777778,
14    "rmse": 0.7737195714912748,
15    "spearman": -1.0,
16    "pearson": -0.9999999999999994
17  },
18  "metrics_real_wom_baseline": {
19    "n": 10,
20    "r2": -0.23456790123456783,
21    "mae": 0.5727511111111111,
22    "rmse": 0.6714200448800041,
23    "spearman": -0.9999999999999999,
24    "pearson": -1.0
25  },
26  "metrics_real_imdb_tuned": {
27    "n": 10,
28    "r2": -0.23456790123456805,
29    "mae": 0.6777777777777778,
30    "rmse": 0.7737195714912748,
31    "spearman": -1.0,
32    "pearson": -0.9999999999999994
33  },
34  "metrics_real_wom_tuned": {
35    "n": 10,
36    "r2": -0.23456790123456783,
37    "mae": 0.5727511111111111,
38    "rmse": 0.6714200448800041,
39    "spearman": -0.9999999999999999,
40    "pearson": -1.0
41  },
42  "metrics_aug_imdb_baseline": {
43    "n": 50,
44    "r2": -0.013460327455919296,
45    "mae": 0.4998999999999998,
46    "rmse": 0.5785758377257038,
47    "spearman": -0.1708965015387121,
48    "pearson": -0.1546915351037204
49  },
50  "metrics_aug_wom_baseline": {
51    "n": 50,
52    "r2": -0.02814640834228621,
53    "mae": 0.46040510000000007,
54    "rmse": 0.5559736496791012,
55    "spearman": -0.21853819706731129,
56    "pearson": -0.22369188468778975
57  },
58  "metrics_aug_imdb_tuned": {
59    "n": 50,
60    "r2": 0.4542151487555801,
61    "mae": 0.3190646385182581,
62    "rmse": 0.4245879408384521,
63    "spearman": 0.6605242616014716,
64    "pearson": 0.6743720308519077
65  },
66  "metrics_aug_wom_tuned": {
67    "n": 50,
68    "r2": 0.06257466661675382,
69    "mae": 0.4306658546709295,
70    "rmse": 0.530878418182961,
71    "spearman": 0.29968787515006,
72    "pearson": 0.25412747776927314

```

73 }
74 }

Baseline classification

ml_dataset/data/model_ready/movie_success_v6/colab_runs_v3/results_classification.json

```

1 {
2   "task": "binary_classification_combined_success",
3   "composite_threshold": 0.11344592183924405,
4   "class_distribution": {
5     "1": 25,
6     "0": 25
7   },
8   "train_size": 40,
9   "test_size": 10,
10  "feature_count": 360,
11  "results_per_model": {
12    "LogReg (ElasticNet)": {
13      "metrics": {
14        "accuracy": 0.9,
15        "f1": 0.8888888888888888,
16        "roc_auc": 0.88,
17        "n_test": 10
18      },
19      "best_params": {
20        "model__C": 0.02310129700083159,
21        "model__l1_ratio": 0.1
22      }
23    },
24    "LogReg (L2)": {
25      "metrics": {
26        "accuracy": 0.9,
27        "f1": 0.8888888888888888,
28        "roc_auc": 0.88,
29        "n_test": 10
30      },
31      "best_params": {
32        "model__C": 0.001
33      }
34    },
35    "Random Forest": {
36      "metrics": {
37        "accuracy": 0.9,
38        "f1": 0.8888888888888888,
39        "roc_auc": 0.8,
40        "n_test": 10
41      },
42      "best_params": {
43        "model__max_depth": 3,
44        "model__max_features": 0.3,
45        "model__min_samples_leaf": 1,
46        "model__n_estimators": 500
47      }
48    },
49    "Gradient Boosting": {
50      "metrics": {
51        "accuracy": 0.8,
52        "f1": 0.8,
53        "roc_auc": 0.7600000000000001,
54        "n_test": 10
55      },
56      "best_params": {
57        "model__l2_regularization": 0.0,
58        "model__learning_rate": 0.05,
59        "model__max_depth": 3,
60        "model__max_iter": 400,
61        "model__min_samples_leaf": 4
62      }
63    },
64    "PLS-DA": {
65      "metrics": {
66        "accuracy": 0.9,
67        "f1": 0.8888888888888888,
68        "roc_auc": 0.88,
69        "n_test": 10
70      },
71      "best_params": {
72        "model__n_components": 1

```

```
73     }
74   },
75   "GP Classifier": {
76     "metrics": {
77       "accuracy": 0.9,
78       "f1": 0.8888888888888888,
79       "roc_auc": 0.8400000000000001,
80       "n_test": 10
81     },
82     "best_params": {
83       "model__kernel": "rbf"
84     }
85   },
86 },
87 "best_model_test": "LogReg (ElasticNet)"
88 }
```

Leakage-free nested cross-validation

ml_dataset/data/model_ready/movie_success_v6/colab_runs_v3/results_nested_cv_finals.json

```

1 {
2   "task": "nested_cv_top3_regressors",
3   "cv_strategy": "outer 5-fold + inner 5-fold (no hyperparameter leakage)",
4   "feature_count": 360,
5   "metrics": {
6     "rf_imdb": {
7       "r2": 0.4921157625589827,
8       "mae": 0.3177409679309426,
9       "rmse": 0.40958051121082134,
10      "spearman": 0.6431027838091413,
11      "pearson": 0.7171847927832931
12    },
13    "rf_wom": {
14      "r2": 0.17373016888768533,
15      "mae": 0.3965547092977455,
16      "rmse": 0.49841106504265614,
17      "spearman": 0.37690276110444176,
18      "pearson": 0.4177022743274261
19    },
20    "gbm_imdb": {
21      "r2": 0.36884438979843237,
22      "mae": 0.34143667045788306,
23      "rmse": 0.456588679964822,
24      "spearman": 0.608933587034129,
25      "pearson": 0.6182542534511072
26    },
27    "gbm_wom": {
28      "r2": 0.0788751508647193,
29      "mae": 0.4174872731786028,
30      "rmse": 0.5262425693493675,
31      "spearman": 0.26137482300026,
32      "pearson": 0.3178387894083348
33    },
34    "pls_imdb": {
35      "r2": 0.4274523447362315,
36      "mae": 0.32868553728237004,
37      "rmse": 0.4348732927235746,
38      "spearman": 0.6485409799155869,
39      "pearson": 0.6589985680158504
40    },
41    "pls_wom": {
42      "r2": 0.1897278534737039,
43      "mae": 0.4020334360923966,
44      "rmse": 0.4935625307425964,
45      "spearman": 0.38525810324129645,
46      "pearson": 0.46090226088852276
47    }
48  },
49  "best_params_per_outer_fold": {
50    "rf_imdb": [
51      {
52        "model__max_depth": 3,
53        "model__max_features": "sqrt",
54        "model__min_samples_leaf": 4,
55        "model__n_estimators": 500
56      },
57      {
58        "model__max_depth": 3,
59        "model__max_features": "sqrt",
60        "model__min_samples_leaf": 1,
61        "model__n_estimators": 200
62      },
63      {
64        "model__max_depth": 5,
65        "model__max_features": "sqrt",
66        "model__min_samples_leaf": 1,
67        "model__n_estimators": 500
68      },
69      {
70        "model__max_depth": 3,
71        "model__max_features": 0.3,
72        "model__min_samples_leaf": 4,

```

```

73     "model__n_estimators": 500
74 },
75 {
76     "model__max_depth": 5,
77     "model__max_features": 0.3,
78     "model__min_samples_leaf": 1,
79     "model__n_estimators": 200
80 }
81 ],
82 "rf_wom": [
83     {
84         "model__max_depth": 3,
85         "model__max_features": 0.3,
86         "model__min_samples_leaf": 1,
87         "model__n_estimators": 200
88     },
89     {
90         "model__max_depth": 5,
91         "model__max_features": "sqrt",
92         "model__min_samples_leaf": 1,
93         "model__n_estimators": 200
94     },
95     {
96         "model__max_depth": 3,
97         "model__max_features": 0.3,
98         "model__min_samples_leaf": 1,
99         "model__n_estimators": 500
100     },
101     {
102         "model__max_depth": 5,
103         "model__max_features": "sqrt",
104         "model__min_samples_leaf": 4,
105         "model__n_estimators": 200
106     },
107     {
108         "model__max_depth": 3,
109         "model__max_features": 0.5,
110         "model__min_samples_leaf": 1,
111         "model__n_estimators": 200
112     }
113 ],
114 "gbm_imdb": [
115     {
116         "model__l2_regularization": 10.0,
117         "model__learning_rate": 0.1,
118         "model__max_depth": 3,
119         "model__max_iter": 100,
120         "model__min_samples_leaf": 10
121     },
122     {
123         "model__l2_regularization": 10.0,
124         "model__learning_rate": 0.1,
125         "model__max_depth": 3,
126         "model__max_iter": 400,
127         "model__min_samples_leaf": 10
128     },
129     {
130         "model__l2_regularization": 10.0,
131         "model__learning_rate": 0.05,
132         "model__max_depth": 3,
133         "model__max_iter": 200,
134         "model__min_samples_leaf": 10
135     },
136     {
137         "model__l2_regularization": 1.0,
138         "model__learning_rate": 0.05,
139         "model__max_depth": 3,
140         "model__max_iter": 400,
141         "model__min_samples_leaf": 10
142     },
143     {
144         "model__l2_regularization": 1.0,
145         "model__learning_rate": 0.1,
146         "model__max_depth": 3,
147         "model__max_iter": 400,
148         "model__min_samples_leaf": 4
149     }
150 ],
151 "gbm_wom": [

```



```

152     {
153         "model__l2_regularization": 10.0,
154         "model__learning_rate": 0.05,
155         "model__max_depth": 3,
156         "model__max_iter": 200,
157         "model__min_samples_leaf": 10
158     },
159     {
160         "model__l2_regularization": 0.0,
161         "model__learning_rate": 0.1,
162         "model__max_depth": null,
163         "model__max_iter": 100,
164         "model__min_samples_leaf": 4
165     },
166     {
167         "model__l2_regularization": 1.0,
168         "model__learning_rate": 0.01,
169         "model__max_depth": 3,
170         "model__max_iter": 100,
171         "model__min_samples_leaf": 4
172     },
173     {
174         "model__l2_regularization": 10.0,
175         "model__learning_rate": 0.01,
176         "model__max_depth": 3,
177         "model__max_iter": 100,
178         "model__min_samples_leaf": 10
179     },
180     {
181         "model__l2_regularization": 0.0,
182         "model__learning_rate": 0.01,
183         "model__max_depth": 3,
184         "model__max_iter": 200,
185         "model__min_samples_leaf": 10
186     }
187 ],
188 "pls_imdb": [
189     {
190         "model__n_components": 1
191     },
192     {
193         "model__n_components": 3
194     },
195     {
196         "model__n_components": 2
197     },
198     {
199         "model__n_components": 3
200     },
201     {
202         "model__n_components": 3
203     }
204 ],
205 "pls_wom": [
206     {
207         "model__n_components": 1
208     },
209     {
210         "model__n_components": 1
211     },
212     {
213         "model__n_components": 1
214     },
215     {
216         "model__n_components": 1
217     },
218     {
219         "model__n_components": 1
220     }
221 ]
222 }
223 }

```

Permutation test and stacking ensemble

ml_dataset/data/model_ready/movie_success_v6/colab_runs_v3/results_stacking.json

```

1 {
2   "task": "stacking_regression_train_test",
3   "train_size": 40,
4   "test_size": 10,
5   "feature_count": 360,
6   "imdb": {
7     "train": {
8       "r2": 0.872666459177191,
9       "mae": 0.1637119529056051,
10      "rmse": 0.20767059118241868,
11      "spearman": 0.9831224397785148,
12      "pearson": 0.9766141745929623
13    },
14    "test": {
15      "r2": 0.6306472395981333,
16      "mae": 0.26186441615121064,
17      "rmse": 0.33092762571165535,
18      "spearman": 0.7416447632896789,
19      "pearson": 0.8328508547779502
20    },
21    "meta_coefs": {
22      "ridge": 0.0268327565337153,
23      "enet": 0.05525907605072579,
24      "rf": 0.7768647770525487,
25      "gbm": -0.08832100813468494,
26      "pls": 0.07903396109930115
27    },
28    "meta_intercept": 1.1417943769592114
29  },
30  "wom": {
31    "train": {
32      "r2": 0.7185373313118816,
33      "mae": 0.24369563534111133,
34      "rmse": 0.2913267357103201,
35      "spearman": 0.9806754221388368,
36      "pearson": 0.9905237544023365
37    },
38    "test": {
39      "r2": 0.01834340970909576,
40      "mae": 0.43790381291960234,
41      "rmse": 0.5383262234665226,
42      "spearman": 0.07878787878787878,
43      "pearson": 0.19728888627420096
44    },
45    "meta_coefs": {
46      "ridge": 0.5470640909454415,
47      "enet": -0.10296195472195226,
48      "rf": 0.36468955031002087,
49      "gbm": -0.1847551996177603,
50      "pls": -0.029687258400116264
51    },
52    "meta_intercept": 0.6895991111965654
53  }
54 }

```

Robust classification

ml_dataset/data/model_ready/movie_success_v6/colab_runs_v3/results_robust_classification.json

```

1 {
2   "task": "robust_classification_sanity_tests",
3   "feature_count": 360,
4   "test_a_permutation": {
5     "real_acc": 0.9,
6     "perm_mean": 0.5065000000000001,
7     "perm_std": 0.15845425207295638,
8     "perm_max": 0.9,
9     "p_value": 0.02,
10    "n_permutations": 200
11  },
12  "test_b_top25_hit": {
13    "LogReg-ElasticNet": {
14      "accuracy": 0.8,
15      "f1": 0.75,
16      "precision": 0.6,
17      "recall": 1.0,
18      "roc_auc": 1.0
19    },
20    "LogReg-L2": {
21      "accuracy": 0.7,
22      "f1": 0.6666666666666666,
23      "precision": 0.5,
24      "recall": 1.0,
25      "roc_auc": 0.9523809523809524
26    },
27    "Random Forest": {
28      "accuracy": 0.7,
29      "f1": 0.0,
30      "precision": 0.0,
31      "recall": 0.0,
32      "roc_auc": 0.8571428571428571
33    },
34    "Gradient Boosting": {
35      "accuracy": 0.6,
36      "f1": 0.5,
37      "precision": 0.4,
38      "recall": 0.6666666666666666,
39      "roc_auc": 0.8095238095238095
40    },
41    "PLS-DA": {
42      "accuracy": 0.9,
43      "f1": 0.8571428571428571,
44      "precision": 0.75,
45      "recall": 1.0,
46      "roc_auc": 0.9047619047619048
47    },
48    "GP Classifier": {
49      "accuracy": 0.7,
50      "f1": 0.0,
51      "precision": 0.0,
52      "recall": 0.0,
53      "roc_auc": 0.9523809523809524
54    }
55  },
56  "test_c_real_only_loo": {
57    "LogReg-ElasticNet": {
58      "accuracy": 0.6,
59      "f1": 0.7142857142857143,
60      "precision": 0.5555555555555556,
61      "recall": 1.0,
62      "roc_auc": 0.6799999999999999
63    },
64    "LogReg-L2": {
65      "accuracy": 0.3,
66      "f1": 0.2222222222222222,
67      "precision": 0.25,
68      "recall": 0.2,
69      "roc_auc": 0.24
70    },
71    "Random Forest": {
72      "accuracy": 0.3,

```

```

73     "f1": 0.2222222222222222,
74     "precision": 0.25,
75     "recall": 0.2,
76     "roc_auc": 0.15999999999999998
77 },
78 "Gradient Boosting": {
79     "accuracy": 0.5,
80     "f1": 0.0,
81     "precision": 0.0,
82     "recall": 0.0,
83     "roc_auc": 0.09999999999999998
84 },
85 "PLS-DA": {
86     "accuracy": 0.3,
87     "f1": 0.2222222222222222,
88     "precision": 0.25,
89     "recall": 0.2,
90     "roc_auc": 0.24
91 },
92 "GP Classifier": {
93     "accuracy": 0.4,
94     "f1": 0.0,
95     "precision": 0.0,
96     "recall": 0.0,
97     "roc_auc": 0.5
98 }
99 },
100 "test_d_synthetic_train_real_test": {
101     "LogReg-ElasticNet": {
102         "accuracy": 0.6,
103         "f1": 0.6,
104         "precision": 0.75,
105         "recall": 0.5,
106         "roc_auc": 0.625
107     },
108     "LogReg-L2": {
109         "accuracy": 0.7,
110         "f1": 0.7272727272727273,
111         "precision": 0.8,
112         "recall": 0.6666666666666666,
113         "roc_auc": 0.7083333333333334
114     },
115     "Random Forest": {
116         "accuracy": 0.6,
117         "f1": 0.5,
118         "precision": 1.0,
119         "recall": 0.3333333333333333,
120         "roc_auc": 0.5416666666666666
121     },
122     "Gradient Boosting": {
123         "accuracy": 0.6,
124         "f1": 0.5,
125         "precision": 1.0,
126         "recall": 0.3333333333333333,
127         "roc_auc": 0.875
128     },
129     "PLS-DA": {
130         "accuracy": 0.8,
131         "f1": 0.8,
132         "precision": 1.0,
133         "recall": 0.6666666666666666,
134         "roc_auc": 0.7291666666666667
135     },
136     "GP Classifier": {
137         "accuracy": 0.6,
138         "f1": 0.6666666666666666,
139         "precision": 0.6666666666666666,
140         "recall": 0.6666666666666666,
141         "roc_auc": 0.7083333333333334
142     }
143 },
144 "best_per_test": {
145     "A_permutation_p_value": 0.02,
146     "B_top25_best": "LogReg-ElasticNet",
147     "C_real_only_best": "LogReg-ElasticNet",
148     "D_synth_to_real_best": "PLS-DA"
149 }
150 }

```

Business classification (synth-to-real)

ml_dataset/data/model_ready/movie_success_v6/colab_runs_v3/results_business_classification.json

```

1 {
2   "task": "business_classification_hit_flop",
3   "thresholds": {
4     "hit_quantile": 0.8,
5     "flop_quantile": 0.3,
6     "hit_threshold": 0.6812225156177218,
7     "flop_threshold": -0.6042711360648163
8   },
9   "class_distributions": {
10    "is_hit": {
11      "0": 40,
12      "1": 10
13    },
14    "is_flop": {
15      "0": 35,
16      "1": 15
17    },
18    "tier": {
19      "average": 25,
20      "flop": 15,
21      "hit": 10
22    }
23  },
24  "feature_count": 360,
25  "hit_detection": {
26    "LogReg-ElasticNet": {
27      "accuracy": 0.6,
28      "f1": 0.5,
29      "precision": 0.3333333333333333,
30      "recall": 1.0,
31      "roc_auc": 0.75,
32      "pr_auc": 0.41666666666666663,
33      "precision_at_recall80": 0.5,
34      "threshold_at_recall80": 0.5634384843845758
35    },
36    "LogReg-L2": {
37      "accuracy": 0.6,
38      "f1": 0.0,
39      "precision": 0.0,
40      "recall": 0.0,
41      "roc_auc": 0.625,
42      "pr_auc": 0.3333333333333333,
43      "precision_at_recall80": 0.3333333333333333,
44      "threshold_at_recall80": 0.1414678636998803
45    },
46    "Random Forest": {
47      "accuracy": 0.8,
48      "f1": 0.0,
49      "precision": 0.0,
50      "recall": 0.0,
51      "roc_auc": 0.625,
52      "pr_auc": 0.325,
53      "precision_at_recall80": 0.4,
54      "threshold_at_recall80": 0.28
55    },
56    "Gradient Boosting": {
57      "accuracy": 0.6,
58      "f1": 0.3333333333333333,
59      "precision": 0.25,
60      "recall": 0.5,
61      "roc_auc": 0.75,
62      "pr_auc": 0.45,
63      "precision_at_recall80": 0.4,
64      "threshold_at_recall80": 0.17672916366877564
65    },
66    "PLS-DA": {
67      "accuracy": 0.7,
68      "f1": 0.0,
69      "precision": 0.0,
70      "recall": 0.0,
71      "roc_auc": 0.6875,
72      "pr_auc": 0.41666666666666663,

```

```

73     "precision_at_recall80": 0.3333333333333333,
74     "threshold_at_recall80": 0.13803184888889824
75 },
76 "GP Classifier": {
77     "accuracy": 0.8,
78     "f1": 0.0,
79     "precision": 0.0,
80     "recall": 0.0,
81     "roc_auc": 0.625,
82     "pr_auc": 0.3333333333333333,
83     "precision_at_recall80": 0.3333333333333333,
84     "threshold_at_recall80": 0.273487564007155
85 }
86 },
87 "flop_avoidance": {
88     "LogReg-ElasticNet": {
89         "accuracy": 0.8,
90         "f1": 0.6666666666666666,
91         "precision": 0.6666666666666666,
92         "recall": 0.6666666666666666,
93         "roc_auc": 0.8095238095238095,
94         "pr_auc": 0.5888888888888889,
95         "precision_at_recall80": 0.6,
96         "threshold_at_recall80": 0.49533173837213734
97     },
98     "LogReg-L2": {
99         "accuracy": 0.8,
100        "f1": 0.6666666666666666,
101        "precision": 0.6666666666666666,
102        "recall": 0.6666666666666666,
103        "roc_auc": 0.8095238095238095,
104        "pr_auc": 0.5888888888888889,
105        "precision_at_recall80": 0.6,
106        "threshold_at_recall80": 0.29033066284006714
107    },
108    "Random Forest": {
109        "accuracy": 0.7,
110        "f1": 0.4,
111        "precision": 0.5,
112        "recall": 0.3333333333333333,
113        "roc_auc": 0.7619047619047619,
114        "pr_auc": 0.5333333333333333,
115        "precision_at_recall80": 0.6,
116        "threshold_at_recall80": 0.314
117    },
118    "Gradient Boosting": {
119        "accuracy": 0.7,
120        "f1": 0.5714285714285714,
121        "precision": 0.5,
122        "recall": 0.6666666666666666,
123        "roc_auc": 0.8095238095238095,
124        "pr_auc": 0.7,
125        "precision_at_recall80": 0.6,
126        "threshold_at_recall80": 0.28001515207970457
127    },
128    "PLS-DA": {
129        "accuracy": 0.7,
130        "f1": 0.4,
131        "precision": 0.5,
132        "recall": 0.3333333333333333,
133        "roc_auc": 0.8095238095238095,
134        "pr_auc": 0.5888888888888888,
135        "precision_at_recall80": 0.6,
136        "threshold_at_recall80": 0.27171311355863725
137    },
138    "GP Classifier": {
139        "accuracy": 0.7,
140        "f1": 0.4,
141        "precision": 0.5,
142        "recall": 0.3333333333333333,
143        "roc_auc": 0.8095238095238095,
144        "pr_auc": 0.5888888888888889,
145        "precision_at_recall80": 0.6,
146        "threshold_at_recall80": 0.3782971798669905
147    }
148 },
149 "synth_to_real_transfer": {
150     "LogReg-ElasticNet": {
151         "hit": {

```

```

152     "accuracy": 0.8,
153     "f1": 0.6666666666666666,
154     "precision": 0.6666666666666666,
155     "recall": 0.6666666666666666,
156     "roc_auc": 0.6666666666666666,
157     "pr_auc": 0.5138888888888888,
158     "precision_at_recall80": 0.375,
159     "threshold_at_recall80": 0.2490442745536388
160 },
161 "flop": {
162     "accuracy": 0.6,
163     "f1": 0.3333333333333333,
164     "precision": 0.3333333333333333,
165     "recall": 0.3333333333333333,
166     "roc_auc": 0.7142857142857142,
167     "pr_auc": 0.6333333333333333,
168     "precision_at_recall80": 0.5,
169     "threshold_at_recall80": 0.37477254337874494
170 }
171 },
172 "LogReg-L2": {
173     "hit": {
174         "accuracy": 0.6,
175         "f1": 0.0,
176         "precision": 0.0,
177         "recall": 0.0,
178         "roc_auc": 0.7619047619047619,
179         "pr_auc": 0.5555555555555556,
180         "precision_at_recall80": 0.5,
181         "threshold_at_recall80": 0.1191353519279237
182     },
183     "flop": {
184         "accuracy": 0.6,
185         "f1": 0.3333333333333333,
186         "precision": 0.3333333333333333,
187         "recall": 0.3333333333333333,
188         "roc_auc": 0.6666666666666666,
189         "pr_auc": 0.4666666666666667,
190         "precision_at_recall80": 0.5,
191         "threshold_at_recall80": 0.149686739846772
192     }
193 },
194 "Random Forest": {
195     "hit": {
196         "accuracy": 0.7,
197         "f1": 0.0,
198         "precision": 0.0,
199         "recall": 0.0,
200         "roc_auc": 0.8095238095238095,
201         "pr_auc": 0.5888888888888889,
202         "precision_at_recall80": 0.6,
203         "threshold_at_recall80": 0.168
204     },
205     "flop": {
206         "accuracy": 0.6,
207         "f1": 0.3333333333333333,
208         "precision": 0.3333333333333333,
209         "recall": 0.3333333333333333,
210         "roc_auc": 0.7142857142857142,
211         "pr_auc": 0.5,
212         "precision_at_recall80": 0.5,
213         "threshold_at_recall80": 0.352
214     }
215 },
216 "Gradient Boosting": {
217     "hit": {
218         "accuracy": 0.8,
219         "f1": 0.5,
220         "precision": 1.0,
221         "recall": 0.3333333333333333,
222         "roc_auc": 1.0,
223         "pr_auc": 1.0,
224         "precision_at_recall80": 1.0,
225         "threshold_at_recall80": 0.4222219096232475
226     },
227     "flop": {
228         "accuracy": 0.7,
229         "f1": 0.6666666666666666,
230         "precision": 0.5,

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231     "recall": 1.0,
232     "roc_auc": 0.5714285714285714,
233     "pr_auc": 0.38333333333333336,
234     "precision_at_recall80": 0.5,
235     "threshold_at_recall80": 0.6716800117444007
236   },
237 },
238 "PLS-DA": {
239   "hit": {
240     "accuracy": 0.6,
241     "f1": 0.0,
242     "precision": 0.0,
243     "recall": 0.0,
244     "roc_auc": 0.6190476190476191,
245     "pr_auc": 0.4206349206349207,
246     "precision_at_recall80": 0.42857142857142855,
247     "threshold_at_recall80": 0.1714737111962889
248   },
249   "flop": {
250     "accuracy": 0.6,
251     "f1": 0.3333333333333333,
252     "precision": 0.3333333333333333,
253     "recall": 0.3333333333333333,
254     "roc_auc": 0.6666666666666667,
255     "pr_auc": 0.41111111111111115,
256     "precision_at_recall80": 0.5,
257     "threshold_at_recall80": 0.24103940111196098
258   }
259 },
260 "GP Classifier": {
261   "hit": {
262     "accuracy": 0.6,
263     "f1": 0.0,
264     "precision": 0.0,
265     "recall": 0.0,
266     "roc_auc": 0.7619047619047619,
267     "pr_auc": 0.5555555555555556,
268     "precision_at_recall80": 0.5,
269     "threshold_at_recall80": 0.32597514506795733
270   },
271   "flop": {
272     "accuracy": 0.6,
273     "f1": 0.3333333333333333,
274     "precision": 0.3333333333333333,
275     "recall": 0.3333333333333333,
276     "roc_auc": 0.7619047619047619,
277     "pr_auc": 0.5333333333333333,
278     "precision_at_recall80": 0.6,
279     "threshold_at_recall80": 0.4012981762568302
280   }
281 }
282 }
283 }

```


Feature-group ablation study

ml_dataset/data/model_ready/movie_success_v6/colab_runs_v3/results_ablation.json

```

1 {
2   "task": "feature_group_ablation",
3   "cv_strategy": "5-fold KFold (random_state=42)",
4   "configs": {
5     "1. Physiology only": {
6       "feature_count": 258,
7       "features_sample": [
8         "eeg_alpha_af7_mean__mean",
9         "eeg_alpha_af7_mean__std",
10        "eeg_alpha_af7_mean_bc__mean",
11        "eeg_alpha_af7_mean_bc__std",
12        "eeg_alpha_af8_mean__mean",
13        "eeg_alpha_af8_mean__std",
14        "eeg_alpha_af8_mean_bc__mean",
15        "eeg_alpha_af8_mean_bc__std",
16        "eeg_alpha_asym_buildup__mean",
17        "eeg_alpha_asym_buildup__std"
18      ]
19    },
20    "2. + Scene": {
21      "feature_count": 267,
22      "features_sample": [
23        "eeg_alpha_af7_mean__mean",
24        "eeg_alpha_af7_mean__std",
25        "eeg_alpha_af7_mean_bc__mean",
26        "eeg_alpha_af7_mean_bc__std",
27        "eeg_alpha_af8_mean__mean",
28        "eeg_alpha_af8_mean__std",
29        "eeg_alpha_af8_mean_bc__mean",
30        "eeg_alpha_af8_mean_bc__std",
31        "eeg_alpha_asym_buildup__mean",
32        "eeg_alpha_asym_buildup__std"
33      ]
34    },
35    "3. + Movie metadata": {
36      "feature_count": 298,
37      "features_sample": [
38        "eeg_alpha_af7_mean__mean",
39        "eeg_alpha_af7_mean__std",
40        "eeg_alpha_af7_mean_bc__mean",
41        "eeg_alpha_af7_mean_bc__std",
42        "eeg_alpha_af8_mean__mean",
43        "eeg_alpha_af8_mean__std",
44        "eeg_alpha_af8_mean_bc__mean",
45        "eeg_alpha_af8_mean_bc__std",
46        "eeg_alpha_asym_buildup__mean",
47        "eeg_alpha_asym_buildup__std"
48      ]
49    },
50    "4. + Self-report (current)": {
51      "feature_count": 340,
52      "features_sample": [
53        "eeg_alpha_af7_mean__mean",
54        "eeg_alpha_af7_mean__std",
55        "eeg_alpha_af7_mean_bc__mean",
56        "eeg_alpha_af7_mean_bc__std",
57        "eeg_alpha_af8_mean__mean",
58        "eeg_alpha_af8_mean__std",
59        "eeg_alpha_af8_mean_bc__mean",
60        "eeg_alpha_af8_mean_bc__std",
61        "eeg_alpha_asym_buildup__mean",
62        "eeg_alpha_asym_buildup__std"
63      ]
64    }
65  },
66  "results": {
67    "1. Physiology only": {
68      "RF_imdb": {
69        "r2": 0.478908360187568,
70        "mae": 0.3244085555555577,
71        "rmse": 0.4148718512945962,
72        "spearman": 0.7056179762186356,

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73     "pearson": 0.7448463185369085,
74     "n_features": 258
75 },
76     "PLS_imdb": {
77         "r2": 0.4847907869668431,
78         "mae": 0.31637604688587356,
79         "rmse": 0.41252353133088526,
80         "spearman": 0.7026823128337404,
81         "pearson": 0.6968801983471019,
82         "n_features": 258
83     },
84     "RF_wom": {
85         "r2": 0.1812200028365395,
86         "mae": 0.39812413142928194,
87         "rmse": 0.49614696550678006,
88         "spearman": 0.40052821128451377,
89         "pearson": 0.4320327039603012,
90         "n_features": 258
91     },
92     "PLS_wom": {
93         "r2": 0.001781468577574663,
94         "mae": 0.43902151338163337,
95         "rmse": 0.5478220911738649,
96         "spearman": 0.2869147659063625,
97         "pearson": 0.30380521014445483,
98         "n_features": 258
99     },
100     "PLS_imdb_tuned": {
101         "r2": 0.4847907869668431,
102         "mae": 0.31637604688587356,
103         "rmse": 0.41252353133088526,
104         "spearman": 0.7026823128337404,
105         "pearson": 0.6968801983471019,
106         "n_features": 258,
107         "best_n_components": 2
108     },
109     "PLS_wom_tuned": {
110         "r2": 0.18378735328058826,
111         "mae": 0.41723477660915376,
112         "rmse": 0.4953685004627598,
113         "spearman": 0.40004801920768307,
114         "pearson": 0.4496045648351638,
115         "n_features": 258,
116         "best_n_components": 1
117     }
118 },
119 "2. + Scene": {
120     "RF_imdb": {
121         "r2": 0.48588407059519956,
122         "mae": 0.3256170285714306,
123         "rmse": 0.41208560754547496,
124         "spearman": 0.6993616444147602,
125         "pearson": 0.7530203816775842,
126         "n_features": 267
127     },
128     "PLS_imdb": {
129         "r2": 0.47889843448919733,
130         "mae": 0.32439974939454225,
131         "rmse": 0.4148758024933247,
132         "spearman": 0.6972441167272946,
133         "pearson": 0.6925609563196331,
134         "n_features": 267
135     },
136     "RF_wom": {
137         "r2": 0.18829908693650454,
138         "mae": 0.39720410164340086,
139         "rmse": 0.49399749264910636,
140         "spearman": 0.41781512605042015,
141         "pearson": 0.4422911546761881,
142         "n_features": 267
143     },
144     "PLS_wom": {
145         "r2": -0.012638823742561422,
146         "mae": 0.4415046155558816,
147         "rmse": 0.5517648294915786,
148         "spearman": 0.284609843937575,
149         "pearson": 0.29279661685828484,
150         "n_features": 267
151     },

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152     "PLS_imdb_tuned": {
153         "r2": 0.47889843448919733,
154         "mae": 0.32439974939454225,
155         "rmse": 0.4148758024933247,
156         "spearman": 0.6972441167272946,
157         "pearson": 0.6925609563196331,
158         "n_features": 267,
159         "best_n_components": 2
160     },
161     "PLS_wom_tuned": {
162         "r2": 0.18241853627015303,
163         "mae": 0.4169986974962383,
164         "rmse": 0.49578370156535556,
165         "spearman": 0.39697478991596635,
166         "pearson": 0.44851109042661236,
167         "n_features": 267,
168         "best_n_components": 1
169     }
170 },
171 "3. + Movie metadata": {
172     "RF_imdb": {
173         "r2": 0.4748745176581407,
174         "mae": 0.3267410000000018,
175         "rmse": 0.41647454582416615,
176         "spearman": 0.6852608350414101,
177         "pearson": 0.740695716029143,
178         "n_features": 298
179     },
180     "PLS_imdb": {
181         "r2": 0.5439217558544576,
182         "mae": 0.2959856615646529,
183         "rmse": 0.38812944793489873,
184         "spearman": 0.7392096654425211,
185         "pearson": 0.7375429622237903,
186         "n_features": 298
187     },
188     "RF_wom": {
189         "r2": 0.21135780510587987,
190         "mae": 0.39183224500651553,
191         "rmse": 0.4869302238438649,
192         "spearman": 0.44076830732292915,
193         "pearson": 0.4731112094530016,
194         "n_features": 298
195     },
196     "PLS_wom": {
197         "r2": 0.021707810475777634,
198         "mae": 0.42793564848354676,
199         "rmse": 0.542326742693162,
200         "spearman": 0.29642256902761105,
201         "pearson": 0.31351779304738137,
202         "n_features": 298
203     },
204     "PLS_imdb_tuned": {
205         "r2": 0.5439217558544576,
206         "mae": 0.2959856615646529,
207         "rmse": 0.38812944793489873,
208         "spearman": 0.7392096654425211,
209         "pearson": 0.7375429622237903,
210         "n_features": 298,
211         "best_n_components": 2
212     },
213     "PLS_wom_tuned": {
214         "r2": 0.09848868451731674,
215         "mae": 0.4041535387372867,
216         "rmse": 0.520609775338064,
217         "spearman": 0.3899639855942376,
218         "pearson": 0.40456370531817976,
219         "n_features": 298,
220         "best_n_components": 10
221     }
222 },
223 "4. + Self-report (current)": {
224     "RF_imdb": {
225         "r2": 0.5425764082728268,
226         "mae": 0.30438409523809645,
227         "rmse": 0.38870148191363024,
228         "spearman": 0.6767425986622873,
229         "pearson": 0.7439411494728424,
230         "n_features": 340

```

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231 },
232 "PLS_imdb": {
233   "r2": 0.46322992422304843,
234   "mae": 0.3108384907089357,
235   "rmse": 0.42106686299141405,
236   "spearman": 0.6679837341368616,
237   "pearson": 0.6842148267002506,
238   "n_features": 340
239 },
240 "RF_wom": {
241   "r2": 0.20127459154856486,
242   "mae": 0.3868244651924295,
243   "rmse": 0.4900331692114709,
244   "spearman": 0.423577430972389,
245   "pearson": 0.45117576659457687,
246   "n_features": 340
247 },
248 "PLS_wom": {
249   "r2": -0.015641098861258307,
250   "mae": 0.428635193880275,
251   "rmse": 0.5525821612789608,
252   "spearman": 0.2711644657863145,
253   "pearson": 0.2842897312369422,
254   "n_features": 340
255 },
256 "PLS_imdb_tuned": {
257   "r2": 0.46322992422304843,
258   "mae": 0.3108384907089357,
259   "rmse": 0.42106686299141405,
260   "spearman": 0.6679837341368616,
261   "pearson": 0.6842148267002506,
262   "n_features": 340,
263   "best_n_components": 2
264 },
265 "PLS_wom_tuned": {
266   "r2": 0.1887742882985196,
267   "mae": 0.40245038382151344,
268   "rmse": 0.4938528687825317,
269   "spearman": 0.37901560624249697,
270   "pearson": 0.4598001596735862,
271   "n_features": 340,
272   "best_n_components": 1
273 }
274 }
275 },
276 "feature_group_sizes": {
277   "physiology": 258,
278   "other": 20,
279   "self_report": 42,
280   "metadata": 31,
281   "scene": 9
282 }
283 }

```

Deep learning (4-layer MLP, n=10,010)

ml_dataset/data/model_ready/movie_success_v7/colab_runs_v7/results_mlp_v7.json

```
1 {
2   "task": "v7_deep_learning_mlp",
3   "architecture": "4-layer MLP (256-128-64-1) with BN, ReLU, Dropout 0.3, AdamW, cosine LR",
4   "n_total": 10010,
5   "n_real": 10,
6   "n_synth": 10000,
7   "n_features": 360,
8   "imdb_5fold_cv": {
9     "r2": 0.6672761459879818,
10    "mae": 0.2822123775210023,
11    "rmse": 0.34401253414878297,
12    "spearman": 0.7516356347139989,
13    "pearson": 0.860775435931638
14  },
15  "wom_5fold_cv": {
16    "r2": 0.2385070086491924,
17    "mae": 0.4506453757516154,
18    "rmse": 0.5739991313038312,
19    "spearman": 0.4764560325559807,
20    "pearson": 0.5157330766880085
21  },
22  "imdb_synth_to_real_transfer": {
23    "r2": 0.7614046335220337,
24    "mae": 0.29282140731811523,
25    "rmse": 0.34013953436387906,
26    "spearman": 0.8231860349807907,
27    "pearson": 0.8727089166641235
28  },
29  "wom_synth_to_real_transfer": {
30    "r2": 0.33097243309020996,
31    "mae": 0.4522523880004883,
32    "rmse": 0.49426381079693216,
33    "spearman": 0.5515151515151515,
34    "pearson": 0.6129603981971741
35  }
36 }
```